

Practice Midterm #1

Name: _____

ID: _____

Score: _____

Problem	Pts.	Score
1	20	
2	20	
3	20	
Total	60	

I certify that this is my sole work and I haven't discussed any problem with other person.

Signature: _____

1. Please print your name and student ID number.
2. Sign on the above statement.
3. For full credit, you should show all of your work and explain your reasoning.

GOOD LUCK!

Problem 1 (20 Pts) (a) (10 Pts) Let $x[n]$ and $y[n]$ be discrete-time signals with DTFTs $X(e^{j\omega})$ and $Y(e^{j\omega})$, respectively. Suppose they satisfy

$$|Y(e^{j\omega})| = |X(e^{j\omega})| \quad \text{and} \quad \angle Y(e^{j\omega}) = -\angle X(e^{j\omega}).$$

Recall that

$$X(e^{j\omega}) = |X(e^{j\omega})|e^{j\angle X(e^{j\omega})} \quad \text{and} \quad Y(e^{j\omega}) = |Y(e^{j\omega})|e^{j\angle Y(e^{j\omega})}.$$

Determine the relationship between $x[n]$ and $y[n]$. In particular, express $y[n]$ in terms of $x[n]$. If you use any DTFT properties, clearly state them.

$$y[n] = x^*[-n] \quad \leftarrow \text{conjugate of } x[n]$$

$$Y(e^{j\omega}) = X^*(e^{j\omega})$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} Y(e^{j\omega}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) e^{j\omega n} d\omega$$

$$= \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega(-n)} d\omega \right)^*$$

(b) (10 Pts) Consider the discrete-time system defined by

$$y[n] = x[n] + 2x[n-1] + x[n-2],$$

IDFT of $X(e^{j\omega})$ at $-n$
 $x[-n]$

where $x[n]$ is the input and $y[n]$ is the output. Determine whether this system is linear, time-invariant, both, or neither. Provide a clear justification for each property you assess.

$$y[n] = x[n] * \delta[n] + x[n] * 2\delta[n-1] + x[n] * \delta[n-2]$$

$$= (\delta[n] + 2\delta[n-1] + \delta[n-2]) * x[n]$$

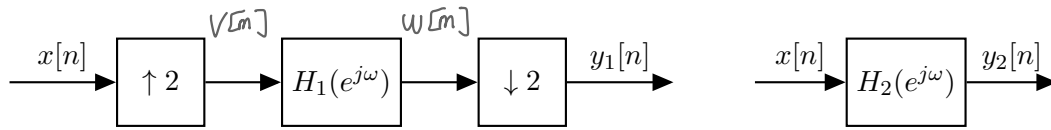
$$x[n] = \delta[n] \quad \Rightarrow \quad h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$$

$$Y(e^{j\omega}) = X(e^{j\omega}) + 2e^{-j\omega} X(e^{j\omega}) + e^{-j2\omega} X(e^{j\omega})$$

$$= (1 + 2e^{-j\omega} + e^{-j2\omega}) X(e^{j\omega})$$

$$h_1[m] \rightarrow \boxed{\downarrow 2} \rightarrow h_2[m]$$

Problem 2 (20 Pts) Consider the system shown below. Suppose that $H_1(e^{j\omega})$ is fixed and known. Find $H_2(e^{j\omega})$, the frequency response of an LTI system, such that $y_2[n] = y_1[n]$ if the inputs to the systems are the same.



$$x[n] \rightarrow \boxed{\uparrow 2} \rightarrow y[m]$$

$$Y(e^{j\omega}) = X(e^{j2\omega})$$

$$x[n] \rightarrow \boxed{\downarrow 2} \rightarrow y[m]$$

$$Y(e^{j\omega}) = \frac{X(e^{j\frac{\omega}{2}}) + X(e^{j(\frac{\omega}{2}-\pi)})}{2}$$

$$V(e^{j\omega}) = X(e^{j2\omega})$$

$$W(e^{j\omega}) = H_1(e^{j\omega}) V(e^{j\omega}) = H_1(e^{j\omega}) X(e^{j2\omega})$$

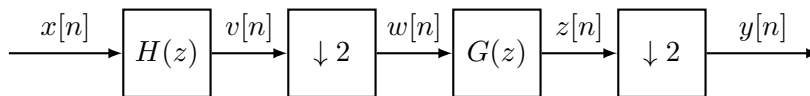
$$Y_1(e^{j\omega}) = \frac{W(e^{j\frac{\omega}{2}}) + W(e^{j(\frac{\omega}{2}-\pi)})}{2}$$

$$W(e^{j\frac{\omega}{2}}) = H_1(e^{j\frac{\omega}{2}}) X(e^{j2 \cdot \frac{\omega}{2}}) = H_1(e^{j\frac{\omega}{2}}) X(e^{j\omega})$$

$$\begin{aligned} W(e^{j(\frac{\omega}{2}-\pi)}) &= H_1(e^{j(\frac{\omega}{2}-\pi)}) X(e^{j2(\frac{\omega}{2}-\pi)}) \\ &= H_1(e^{j(\frac{\omega}{2}-\pi)}) X(e^{j\omega} e^{-j2\pi}) \\ &= H_1(e^{j(\frac{\omega}{2}-\pi)}) X(e^{j\omega}) \end{aligned}$$

$$Y_1(e^{j\omega}) = \frac{H_1(e^{j\frac{\omega}{2}}) + H_1(e^{j(\frac{\omega}{2}-\pi)})}{2} X(e^{j\omega})$$

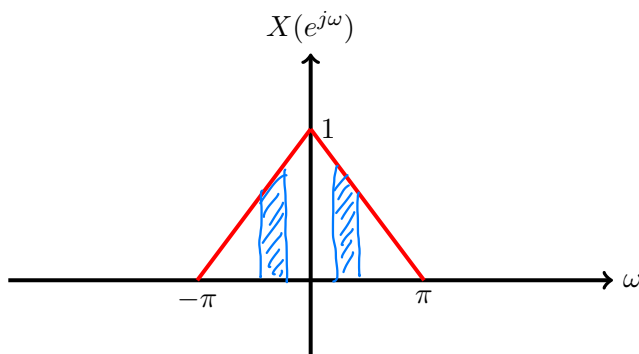
Problem 3 (20 Pts) We consider the following system.



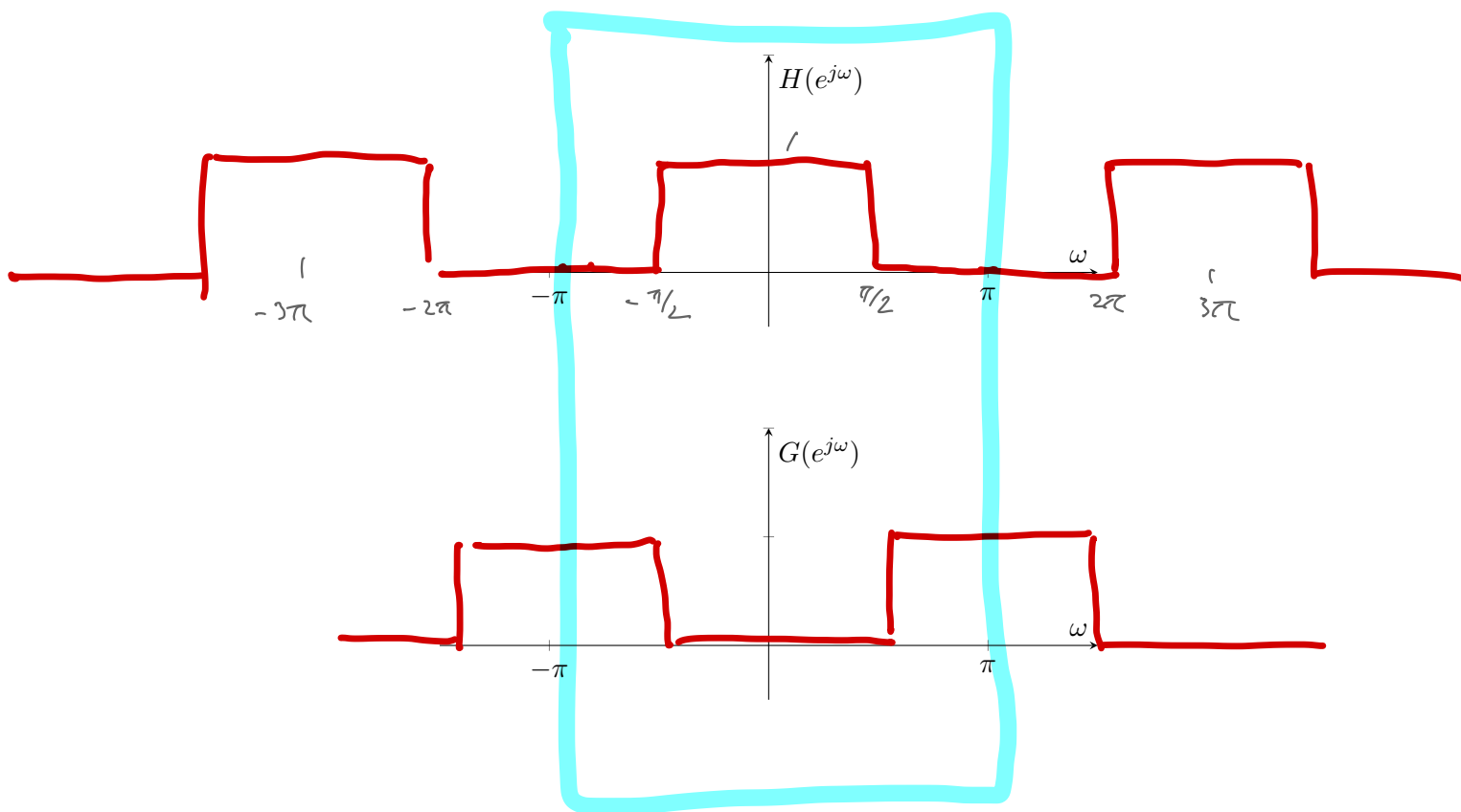
Let $H(z)$ be an ideal lowpass filter satisfying

$$H(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \frac{\pi}{2}, \\ 0 & \frac{\pi}{2} < |\omega| \leq \pi, \end{cases}$$

and the second filter $G(z)$ be given from $H(z)$ by $G(e^{j\omega}) = H(e^{j(\omega-\pi)})$. Suppose that the DTFT of $x[n]$ is given as in the following plot.

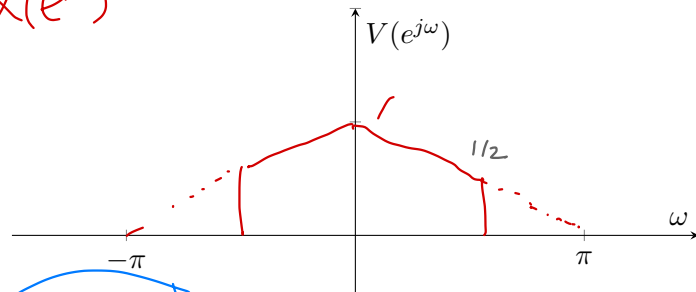


(a) (10 Pts) Draw $|H(e^{j\omega})|$ and $|G(e^{j\omega})|$ for $\omega \in [-\pi, \pi]$.



(b) (20 Pts) Draw the DTFTs of $v[n]$, $w[n]$, $z[n]$, and $y[n]$. Label all graphs. Recall that all DTFTs are 2π -periodic.

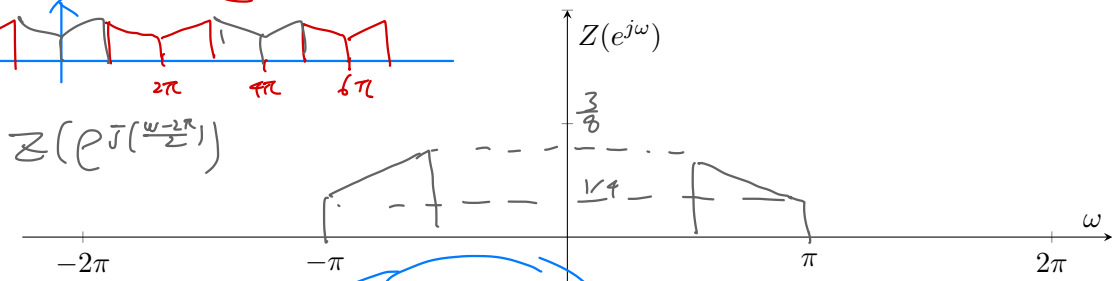
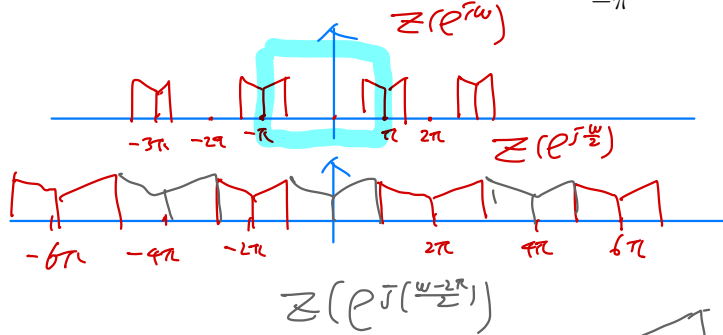
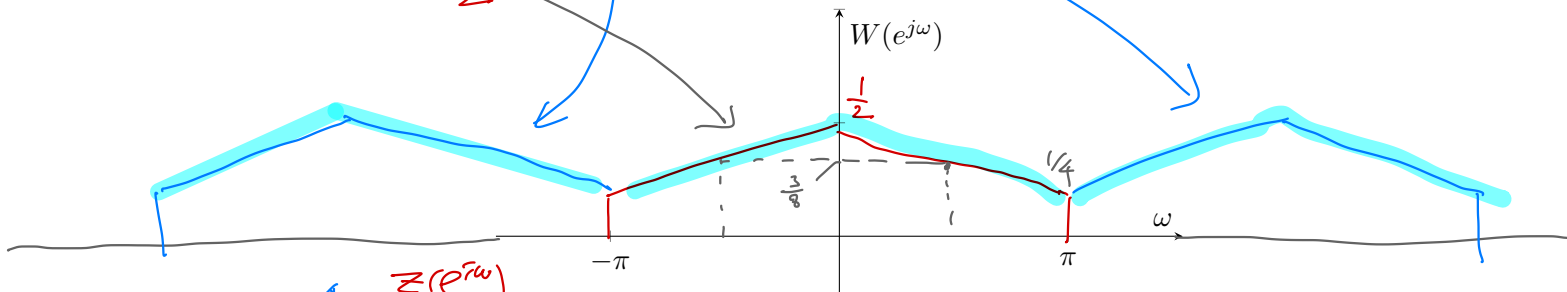
$$V(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega})$$



$$x(t) = \begin{cases} 1 & t=0 \\ 0 & t \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = 0$$

$$W(e^{j\omega}) = \frac{V(e^{j\frac{\omega}{2}}) + V(e^{j(\frac{\omega}{2}-\pi)})}{2}$$



$$Y(e^{j\omega}) = \frac{1}{2} Z(e^{j\frac{\omega}{2}}) + \frac{1}{2} Z(e^{j(\frac{\omega}{2}-\pi)})$$

$$e^{j(\frac{\omega}{2}-\pi)} = e^{j(\frac{\omega-2\pi}{2})}$$

$$e^{j\frac{\omega}{2}}$$

